

Data Quality & Profiling Report

Auto-generated by `etl/profile_data.py`. The dataset is synthetic but representative of a regional surveillance context. Issues found here are handled explicitly in `etl/load.py`.

countries.csv

- Rows: **20**, Columns: **6**
- Columns: `iso3`, `country_name`, `afro_subregion`, `latitude`, `longitude`, `priority_country`
- Missing values: none
- Fully duplicated rows: 0

population.csv

- Rows: **100**, Columns: **5**
- Columns: `iso3`, `year`, `total_population`, `under5_population`, `urban_population_pct`
- **Missing values:**
 - `total_population` : 1 (1.0%)
 - `under5_population` : 8 (8.0%)
 - `urban_population_pct` : 8 (8.0%)
- Fully duplicated rows: 0

disease_surveillance.csv

- Rows: **700**, Columns: **7**
- Columns: `iso3`, `year`, `disease`, `cases_reported`, `deaths_reported`, `attack_rate_per_100k`, `case_fatality_ratio_pct`
- Missing values: none
- Fully duplicated rows: 0

outbreaks.csv

- Rows: **386**, Columns: **9**
- Columns: `outbreak_id`, `iso3`, `year`, `disease`, `start_date`, `duration_days`, `time_to_detection_days`, `cases`, `deaths`
- Missing values: none
- Fully duplicated rows: 0

laboratory_capacity.csv

- Rows: **100**, Columns: **7**
- Columns: `iso3`, `year`, `total_public_labs`, `labs_iso15189_accredited`, `iso15189_accreditation_pct`, `avg_turnaround_time_days`, `diagnostic_tests_per_100k`
- Missing values: none
- Fully duplicated rows: 0

reporting_metrics.csv

- Rows: **100**, Columns: **5**
- Columns: iso3, year, timeliness_pct, completeness_pct, idsr_weekly_compliance_pct
- Missing values: none
- Fully duplicated rows: 0

workforce.csv

- Rows: **100**, Columns: **8**
- Columns: iso3, year, epidemiologists_total, epidemiologists_per_100k, feltp_trained_total, feltp_trained_pct, lab_technicians_total, lab_technicians_per_100k
- Missing values: none
- Fully duplicated rows: 0

funding.csv

- Rows: **100**, Columns: **7**
- Columns: iso3, year, total_funding_usd, domestic_funding_usd, external_funding_usd, funding_per_capita_usd, domestic_funding_share_pct
- **Missing values:**
 - total_funding_usd : 6 (6.0%)
 - domestic_funding_usd : 2 (2.0%)
 - external_funding_usd : 3 (3.0%)
 - funding_per_capita_usd : 7 (7.0%)
 - domestic_funding_share_pct : 7 (7.0%)
- Fully duplicated rows: 0

Cross-cutting checks

- Reference countries: **20** (expected 20)
- priority_country values: [np.int64(1), np.int64(2), np.int64(3)]

Referential integrity (iso3 in countries.csv?)

- population.csv : OK
- disease_surveillance.csv : OK
- outbreaks.csv : OK
- laboratory_capacity.csv : OK
- reporting_metrics.csv : OK
- workforce.csv : OK
- funding.csv : OK

Panel completeness (expected 20 countries x 5 years = 100 rows)

- population.csv : 100 country-year rows, years=[np.int64(2021), np.int64(2022), np.int64(2023), np.int64(2024), np.int64(2025)]
- laboratory_capacity.csv : 100 country-year rows, years=[np.int64(2021), np.int64(2022), np.int64(2023), np.int64(2024), np.int64(2025)]

- `reporting_metrics.csv` : 100 country-year rows, years=[np.int64(2021), np.int64(2022), np.int64(2023), np.int64(2024), np.int64(2025)]
- `workforce.csv` : 100 country-year rows, years=[np.int64(2021), np.int64(2022), np.int64(2023), np.int64(2024), np.int64(2025)]
- `funding.csv` : 100 country-year rows, years=[np.int64(2021), np.int64(2022), np.int64(2023), np.int64(2024), np.int64(2025)]

Duplicate (iso3, year) keys

- `population.csv` : 0
- `laboratory_capacity.csv` : 0
- `reporting_metrics.csv` : 0
- `workforce.csv` : 0
- `funding.csv` : 0

Numeric range / plausibility checks

population.csv

- Numeric ranges:
 - `total_population` : min=5.49e+06, max=2.35e+08, mean=4.91e+07, neg=0
 - `under5_population` : min=7.82e+05, max=3.97e+07, mean=7.71e+06, neg=0
 - `urban_population_pct` : min=25.2, max=64, mean=46.6, neg=0
- `urban_population_pct` outside 0-100: 0
- `under5 > total population`: 0
- `under5 share range`: 13.0% - 18.0% (implausibly wide swings suggest noisy under5 figures)

reporting_metrics.csv (percentages should be 0-100)

- Numeric ranges:
 - `timeliness_pct` : min=51.5, max=98.4, mean=72.6, neg=0
 - `completeness_pct` : min=62, max=99.5, mean=81.4, neg=0
 - `idsr_weekly_compliance_pct` : min=54.5, max=100, mean=76.1, neg=0
- `timeliness_pct` outside 0-100: 0
- `completeness_pct` outside 0-100: 0
- `idsr_weekly_compliance_pct` outside 0-100: 0

funding.csv

- Numeric ranges:
 - `total_funding_usd` : min=5.05e+06, max=5.07e+08, mean=1.06e+08, neg=0
 - `domestic_funding_usd` : min=7.83e+05, max=2.04e+08, mean=4.79e+07, neg=0
 - `external_funding_usd` : min=3.91e+06, max=3.03e+08, mean=5.78e+07, neg=0
 - `funding_per_capita_usd` : min=0.75, max=5.08, mean=2.16, neg=0
 - `domestic_funding_share_pct` : min=13.3, max=63.8, mean=36.6, neg=0
- rows where `|domestic+external - total| > 1%`: 0

laboratory_capacity.csv

- Numeric ranges:
 - `total_public_labs` : min=2, max=44, mean=12, neg=0

- labs_iso15189_accredited : min=0, max=16, mean=3.08, neg=0
 - iso15189_accreditation_pct : min=2.3, max=55.6, mean=25, neg=0
 - avg_turnaround_time_days : min=1.2, max=11.5, mean=6.33, neg=0
 - diagnostic_tests_per_100k : min=26, max=234, mean=95.8, neg=0
- accredited labs > total labs: 0

disease_surveillance.csv

- diseases tracked: ['Cholera', 'Lassa fever', 'Measles', 'Meningitis', 'Polio (cVDPV)', 'Viral haemorrhagic fever', 'Yellow fever']
- Numeric ranges:
 - cases_reported : min=0, max=2.06e+05, mean=9.25e+03, neg=0
 - deaths_reported : min=0, max=1e+04, mean=436, neg=0
 - attack_rate_per_100k : min=0, max=132, mean=16.7, neg=0
 - case_fatality_ratio_pct : min=0.13, max=125, mean=22.8, neg=0
- deaths > cases: 7

outbreaks.csv

- diseases: ['Cholera', 'Lassa fever', 'Measles', 'Meningitis', 'Polio (cVDPV)', 'Viral haemorrhagic fever', 'Yellow fever']
- Numeric ranges:
 - duration_days : min=15, max=190, mean=63.5, neg=0
 - time_to_detection_days : min=0, max=26, mean=10.7, neg=0
 - cases : min=6, max=717, mean=178, neg=0
 - deaths : min=0, max=409, mean=23.1, neg=0
- distinct outbreak_id: 386 of 386 rows
- year range: 2021-2025